PSEUDO-COHERENCE OVER A NOETHERIAN RING VIA FINITENESS OF COHOMOLOGY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Editorial note. Benign-screen note: the proof contains the phrase 'it is clear that' / 'is obvious' once, used as a legitimate step inside a complete argument; the author proves the statement, so this is not a hint-only proof.

Allowed standard prerequisites (editorial).

- Finite modules over a Noetherian ring are pseudo-coherent, and pseudo-coherent modules are finite. Algebra, Lemma \ref{algebra-lemma-resolution-by-finite-free} and Lemma \ref{lemma-n-pseudo-module}, cited at proof lines 16531-16535. Settle the module statement.
- Cohomology-by-cohomology criterion for m-pseudo-coherence. Lemma \ref{lemma-cohomology-pseudo-coherent}, cited at proof lines 16543-16545. Gives the converse implication.

Source reference. Original lemma 066E: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7933–7933. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 00LP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 795–795. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 064T: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7877–7877. Navigation pointer to a statement the

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Source reference. Original lemma 066B: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7930–7930. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 064U: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7878–7878. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let R be a Noetherian ring. Then*

(1) *A complex of R-modules $K^\bullet$ is m-pseudo-coherent if and only if $K^\bullet \in D^-(R)$ and $H^i(K^\bullet)$ is a finite R-module for $i \geq m$.*
(2) *A complex of R-modules $K^\bullet$ is pseudo-coherent if and only if $K^\bullet \in D^-(R)$ and $H^i(K^\bullet)$ is a finite R-module for all i.*
(3) *An R-module is pseudo-coherent if and only if it is finite.*

Proof. In Algebra, Lemma 00LP we have seen that any finite R-module is pseudo-coherent. On the other hand, a pseudo-coherent module is finite, see Lemma 064T. Hence (3) holds. Suppose that $K^\bullet$ is an m-pseudo-coherent complex. Then there exists a bounded complex of finite free R-modules $E^\bullet$ such that $H^i(K^\bullet)$ is isomorphic to $H^i(E^\bullet)$ for $i > m$ and such that $H^m(K^\bullet)$ is a quotient of $H^m(E^\bullet)$. Thus it is clear that each $H^i(K^\bullet)$, $i \geq m$ is a finite module. The converse implication in (1) follows from Lemma 066B and part (3). Part (2) follows from (1) and Lemma 064U. □